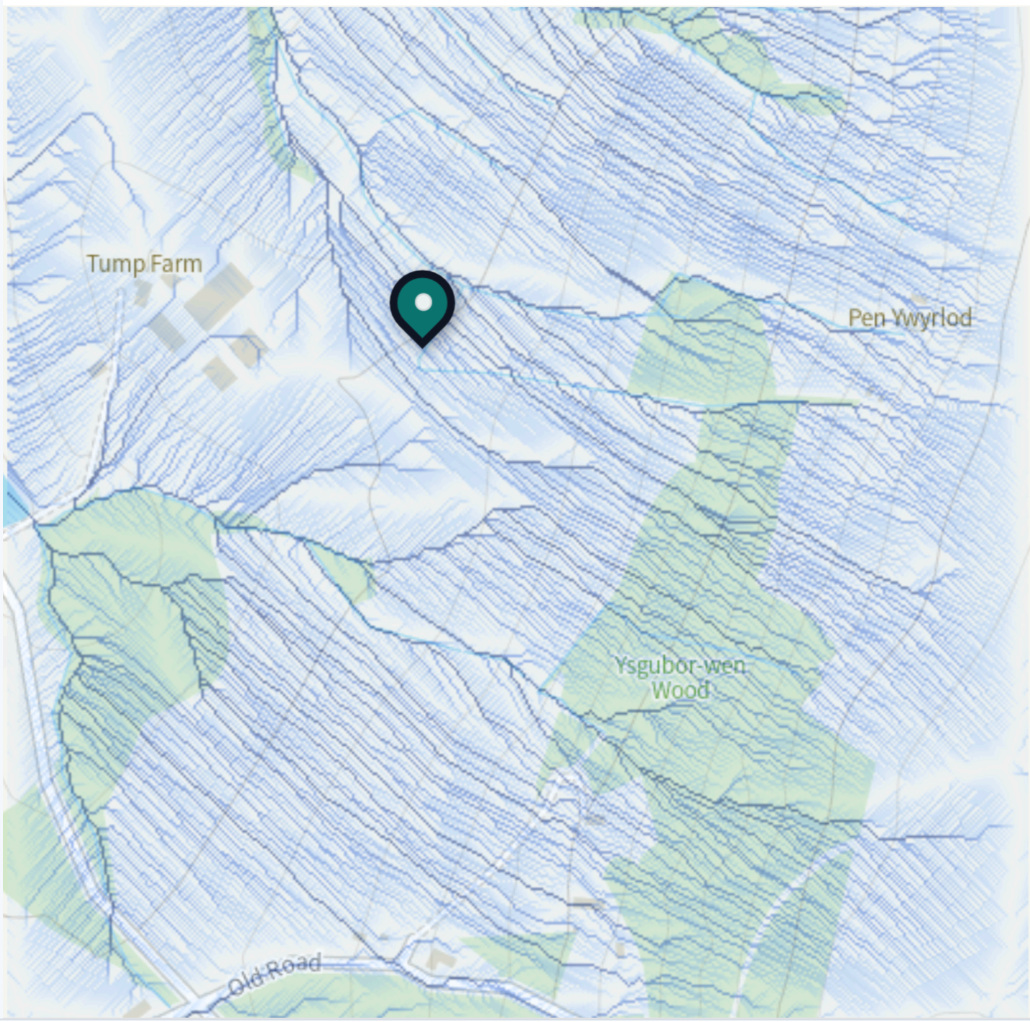
Tump Farm Spring Recharge - Pre-Visit Desk Study

Prepared for discussion before the site visit. Generated 15 August 2026.

> This note uses LiDAR-derived surface mapping and a simple water-balance scenario to show where there may be an opportunity to hold more rainfall and runoff on the land. The figures below describe opportunity, not a guaranteed increase in spring flow.

LiDAR contours and modelled surface-flow concentration



Blue lines show modelled surface-flow concentration from public LiDAR terrain data. They are useful field-check targets: places to look for wet flushes, compacted flow lines, drains, or evidence that the model is missing something.

Worked Opportunity Scenario

RAINFALL USED	RECHARGE STRIP	UPSLOPE AREA	OPPORTUNITY
900 mm/year	**1 ha**	**3 ha**	**7.57 ML/year**

The scenario asks: if one hectare of strategically placed broadleaf shelterbelt or recharge strip intercepted runoff from three hectares of compacted upslope pasture, how much extra water might be retained or infiltrated on site?

Using 900 mm annual rainfall, the arithmetic gives an indicative opportunity of **7.57 million litres per year** compared with the same hectare remaining as compacted pasture. This should be treated as a scoping number for discussion, not as a measured aquifer recharge or spring-yield prediction.

WATER MOVEMENT	PASTURE BASELINE	RECHARGE STRIP SCENARIO	OPPORTUNITY SHOWN
Rain falling directly on 1 ha	9.00 ML/year	9.00 ML/year	Same rainfall input
Upslope runoff captured from 3 ha	0 ML/year	5.67 ML/year	Water that might otherwise leave quickly as surface runoff
Runoff from the 1 ha feature itself	-2.70 ML/year	-0.18 ML/year	2.52 ML/year retained on the feature area
Vegetation water use and interception	-4.50 ML/year grass evapotranspiration assumption	-5.12 ML/year broadleaf interception/transpiration assumption	-0.62 ML/year additional tree water cost
Net retained / infiltrated opportunity	**1.80 ML/year**	**9.37 ML/year**	**+7.57 ML/year**

Cost Sensitivity and Staging

One hectare is a large intervention. It can be staged: current Welsh Small Woodland Creation guidance supports planting totals from **0.1 ha to 1.99 ha**, made up from blocks as small as **0.01 ha**. The main cost trap for narrow shelterbelts is fencing.

EXAMPLE SHAPE	AREA	APPROX. FENCE LENGTH	GRANT-RATE FENCE COST	WHAT IT MEANS
10 m wide strip	1 ha: 10 m x 1,000 m	About 2,020 m perimeter	About £16.8k stock fence or £26.9k deer fence, before gates	Hydrologically interesting, but fence-heavy.
25 m wide strip/block	1 ha: 25 m x 400 m	About 850 m perimeter	About £7.1k stock fence or £11.3k deer fence, before gates	Same area, much less fence per litre of opportunity.
0.25 ha trial	10 m x 250 m	About 520 m perimeter	About £4.3k stock fence or £6.9k deer fence, before gates	Smaller proof-of-concept before committing to the full hectare.

Current Welsh guidance gives indicative capital rates of **£8.33/m** for post and wire fencing with stock netting and **£13.32/m** for deer fencing. Funded fencing must be needed to make the new woodland stockproof, and the funded length must be no longer than the perimeter of the planting area. Field gates are capped at five per application.

Litres per Pound Heuristic

The table below is a scoping heuristic for choosing what to inspect on the visit. It compares rough water-retention opportunity with rough capital cost. It should not be read as a grant claim, engineering design or promised spring-flow increase.

INTERVENTION	INDICATIVE ANNUAL WATER OPPORTUNITY	ROUGH COST SENSITIVITY	INDICATIVE LITRES/YEAR PER £	BEST USE
Broadleaf recharge strip / shelterbelt	Up to 7.57 ML/year in the worked 1 ha scenario	High where narrow strips need long new fences. Much better if existing boundaries reduce fencing.	Roughly 250-650 L/year/£ using £12k-£30k first-pass capital cost range	Durable, multi-benefit intervention where layout and funding work.
Contour swale or infiltration bund	Potentially hundreds of thousands to a few million litres/year, depending on upslope area intercepted	Often lower than fencing if simple earthworks are possible, but needs careful siting and overflow design.	Could be competitive with tree strips, but uncertainty is high until surveyed	Good staged trial for slowing runoff and testing infiltration response.
Leaky woody check dams / small flow-path blocks	Small storage per structure, but repeated many times during wet periods	Low cost where timber/material is available and the flow path is suitable.	Potentially high, but very site-specific	Useful first intervention on the darkest blue flow lines if permissions and channel conditions fit.
Pond or scrape	Visible storage volume can be measured, but not all stored water becomes recharge	Can become expensive once design, spoil, ecology, safety and permissions are included.	Wide range; often better justified by biodiversity, stock water or amenity as well as recharge	Worth considering after the visit, but should not be assumed to be the best recharge option.

> Practical recommendation: use the visit to find the cheapest high-confidence first experiment. Look for existing fences, natural flow concentration, wet flushes, compacted tracks, and places where a small intervention can be monitored before scaling up.

Assumptions

ASSUMPTION	VALUE USED	WHY IT IS REASONABLE FOR A FIRST PASS
Annual rainfall	900 mm/year	Simple conservative round number. Met Office Usk long-term average is about 995 mm/year for 1981-2010.
Pasture runoff	30% of rainfall	Scenario value representing compacted pasture and rapid surface runoff. Needs field observation/infiltration testing.
Runoff after tree/recharge feature	2% of rainfall on the feature	Optimistic scenario inspired by Pontbren evidence that tree shelterbelts can greatly increase infiltration and reduce runoff.
Upslope capture efficiency	70% of runoff from 3 ha	Design-target scenario for a well-placed contour feature. Actual performance depends on layout, soil, slope and maintenance.
Broadleaf tree water cost	5.12 ML/year over 1 ha	Combines 18% rainfall interception and 350 mm/year transpiration. Forestry Research gives broadleaf interception and transpiration ranges consistent with this order of magnitude.

Evidence Base

- **Local rainfall:** Met Office long-term averages for Usk give annual rainfall around 995 mm for 1981-2010. This note uses 900 mm to keep the worked example simple and slightly conservative.
- **LiDAR terrain:** the map uses DataMapWales public LiDAR DTM data to show surface contours and modelled surface-flow concentration.
- **Welsh shelterbelt evidence:** Pontbren research found much higher infiltration under tree shelterbelts than adjacent grazed pasture, with later summaries reporting infiltration around 60-67 times faster and runoff volumes reduced by about 78% under trees compared with grassland.
- **Tree water use:** Forestry Research summarises that broadleaves generally intercept around 10-25% of annual rainfall and transpire roughly 300-390 mm/year.

Interpretation

> The important conclusion is not that a particular intervention will add exactly 7.57 ML/year to the spring. The useful conclusion is that the land may contain a sizeable water-retention opportunity if surface runoff can be slowed and encouraged to infiltrate in the right places.

The LiDAR map also shows why a site visit matters. Surface-flow lines can help choose where to walk and what to inspect, but spring flow may be controlled by deeper geology, fractures, perched groundwater, drains, or old land management features.

Limitations

> This is a pre-visit desk study. It is suitable for scoping, discussion and planning field observations. It is not a hydrogeological assessment and should not be used as a guaranteed prediction of spring yield.

- The blue flow paths are modelled from ground surface shape, not measured groundwater movement.
- The spring's real recharge area may extend outside the visible surface catchment.
- The calculation is sensitive to runoff percentage, capture efficiency and soil infiltration.
- Field observations should check whether mapped flow lines match wet ground, drains or erosion evidence.

Source Links

- Met Office: Usk location-specific long-term averages
- DataMapWales: LiDAR Data Download
- Carroll et al. 2004: Can tree shelterbelts on agricultural land reduce flood risk?
- Pontbren summary: tree planting and reduced runoff
- The Pontbren Project report
- Forest Research: forestry and water resources
- Forestry Commission Information Note: Water Use by Trees
- Welsh Government: Small Grants - Woodland Creation window 10 rules
- Welsh Government: SFS Optional Small Woodland Creation technical guidance
- Welsh Government: Woodland Creation Grant window 6 rules
- Innovative Farmers: shelterbelt design and cost example
- Davis et al. 2023: organic shelterbelt costs and design
- Welsh Government: natural flood management guidance

Spring coordinate used for mapping: 51.664158, -2.855463. Static map screenshot prepared from the JobDone Tumptonics Water Walk map.